Update: the Calibration Meets the Routing Theory

Hossein Piri

September 2026

This note answers the meeting's two open empirical questions, what flips the myopia result (Meichun) and whether the flat parameters are an estimation artifact (Xinyuan), and adds two initiatives of my own: computing the routing theory directly at the calibrated primitives, and a numerical audit of two properties that Xinyuan's August 19 notes describe as numerically observed but unproven, taken up under his invitation to stress-test the drafts. Code and full results: `github.com/hosseinpiri/recommendation-systems`, folder `phase1_2_Aug2026`.

1 Boosting ρ, and the real gate

Raising ρ counterfactually, Meichun's suggestion, does flip the result, monotonically: the aware-minus-myopic gap runs from 0.04% at the calibrated $\rho = 0.035$ (statistically zero) to 0.99% at $\rho = 1$, where the primitive route used as a policy captures 0.89%, nearly the full premium, so the route is a near-optimal policy wherever cultivation has value. But one percent at $\rho = 1$ is a cap, and the phase map explains it: crossing intercept-spread multipliers $\{1, 3, 6\}$ with $\kappa \in \{1.966, 4, 6\}$ at $\rho = 0.45$, operationalizing Xinyuan's α and κ perturbations, a sixfold intercept spread moves the gap from 0.44% only to 0.76%, while $\kappa = 6$ at the original spread moves it to 21.2%. The near-uniform consumption probabilities (average 4%, nearly common across items) are a symptom of small κ, not of close intercepts; alignment discrimination is the single gate behind every null so far. Two loose ends from the same discussion close the same way: removing α entirely, refitting with a single common intercept, gives $\kappa = 1.34$ and simulated gaps of 0.03% at $\hat{\rho}$ and 1.12% at $\rho = 0.45$, so nothing was driven by the intercepts; and the general-σ question is answered by the spline diagnostics, where freeing the functional form improves holdout log loss only in the fourth decimal, so the logistic is not the constraint either.

The sweep also extends past full adoption. For $\rho \in (1, 2)$ the transition is well defined, $T_i(z) - x_i = (1 - \rho)(z - x_i)$ places the state on the far side of the consumed item while contracting at rate $\rho - 1$ (reflection at $\rho = 2$, divergence beyond), and the tube machinery carries over with radius $|1 - \rho|D$. In the showcase this overshoot regime is a slingshot: the exact-DP premium rises from 169% at $\rho = 1$ to 254% at $\rho = 1.95$ and the optimal route shortens from two bridges to one. The data want none of it: in the category simplex the projected transition saturates at the clicked vertex for every $\rho \geq 1$, making overshoot empirically indistinguishable from full adoption, and the entire region is rejected against $\hat{\rho} = 0.035$ by 6,631 log-likelihood points. Overshooting belongs in the theory's parameter map, not in the news calibration.

2 The estimation is not forcing the data

Regenerating click labels on the real design matrix from known truths (intercept spreads up to six times ours, κ up to 6, all recentred to the 4% click rate), the production estimation recovers κ to within 0.022 and the intercept profile with correlation at least 0.995 in all nine configurations. If tastes here were heterogeneous and alignment-sensitive, the pipeline would have said so; the flat calibration is a property of news reading, not of the method.

3 The routing theory predicts our null

The $\rho = 1$ theory is directly computable on the 14-category graph. The skip inequality $1/p_{j\to k} + 1/p_{k\to i} - m_k/\bar{r}_{a^\star} \leq 1/p_{j\to i}$ holds for none of the 2,184 ordered triples at calibration (minimum slack 17): no stepping stone pays, so the optimal route is direct harvesting, which is myopia. The dense content Gram admits 51 profitable triples, but the turnpike threshold is $\bar{H} \approx 167$ (one-hot) and ≈ 137 (dense) against 30-period sessions, and patience breakpoints start at retention $\delta \approx 0.60$ in the $\rho = 1$ idealization: sessions this short play myopically by the theory's own account. Xinyuan's artificial-item test asked whether breaking the uniformity of consumption probabilities produces a nontrivial policy: inserting an $\alpha = -10$, margin-5 category makes the probabilities sharply non-uniform, yet the skip inequality still holds for no triple and the optimal route stays myopic; non-uniformity alone is not enough at our κ, exactly as the phase map predicts. (Frontier notes: one-hot features make the full dominance frontier structurally 100%; the meaningful (α, m)-restricted frontier is {music, lifestyle, health, finance}, and the artificial item joins it on margin alone, confirming the documented non-tightness.)

4 The two conjectured properties fail, in scope

Xinyuan's two August 19 notes, *Optimal Path* (routing structure for the discounted and finite-time settings) and *Efficient Frontier* (dominance and route characterization), describe two properties as numerically observed but unproven: the ordered-marginals inequality of Remark 7 in *Optimal Path* ("one lemma would settle three things") and the single-crossing property behind Conjecture 1 in *Efficient Frontier*. Over 2,000 random instances (exact node recursion, $H = 60$), 1951 satisfy Assumption 2 in both stated forms. Within scope, the ordered-marginals inequality fails in 540 instances in its upper half and 1606 in its full two-sided form; single crossing fails at a comparable rate, with up to 20 sign changes in one instance. The violations occur deep in the turnpike regime, sit nine orders above float noise, and an independent audit reproduced every count and verified the saved examples with separate code. Three complete counterexample instances with full primitives, each certified in scope, are in `output/p25_ordered_marginals.json` in the repository above, ready for direct reconstruction. The constructive reading: the exact ladder and the interval conjecture need a primitive condition defining a subclass, and that search is now example-guided rather than open-ended.

5 Why the Table 1 intercepts look uniform

The concern was that the category intercepts are almost the same constant across items, and that after the sigmoid even their differences stop mattering. Both halves now have answers. The flatness is largely aggregation: refitting with subcategory intercepts (122 subcategories with sufficient data), the spread is 4.2 log-odds (1.8 between the 5th and 95th percentiles), five times the category-level 0.85. The world is not flat; averaging heterogeneous article types into 14 categories flattened it, which argues for a finer action space in the next dataset round. And the second half of the concern is exactly what the phase map in Section 1 establishes: even a sixfold intercept spread buys almost nothing at our κ, because with weak alignment discrimination the sigmoid compresses whatever differences exist. The two observations are one fact: consumption probabilities are near-uniform because κ is small, and the intercepts merely look uniform because of aggregation.

A separate new fact surfaced by the same round, not raised in the meeting but worth the group's attention: the transition is not one ρ. Per-category profiling gives $\rho_c \approx 0.05$ for news, sports, and food, 0.02 for finance and entertainment, and $\rho_c \approx 0$ for travel and health (likelihood ratio 92 against the common value for travel alone). The pattern reads as recency-driven versus need-driven content and strengthens the persistent-interest interpretation of ρ.

6 Resistance and the index, revised

Resistance: exists, and small, which is what the theory needs. On the full impression scale, jointly estimated with ρ, backfire is decisively present (the simplex-native proportional-redistribution form is data-preferred, likelihood ratio 1,489) with a per-impression magnitude of 0.002 to 0.01, attention-diluted, far too small to threaten the routing structure. Its policy value is material only when paired with $\kappa \geq 5$. Proposal: adopt it as a perturbation term in the proportional form, keep the no-resistance case as the solved baseline, and cite the measured magnitude as the smallness justification.

Index policy: the route is the missing potential. The route value W supplies a potential with a certified error bound where the earlier geometric guess reached 97% of oracle and the myopic-payoff potential failed. The benchmark to beat remains estimate-then-plug-in planning, near-oracle whenever a few persistent intercepts are the only unknowns, so the index's case should be built where plug-in struggles: state uncertainty or short-lived items, on the route-based Φ.